PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date: Aug 04, 2026

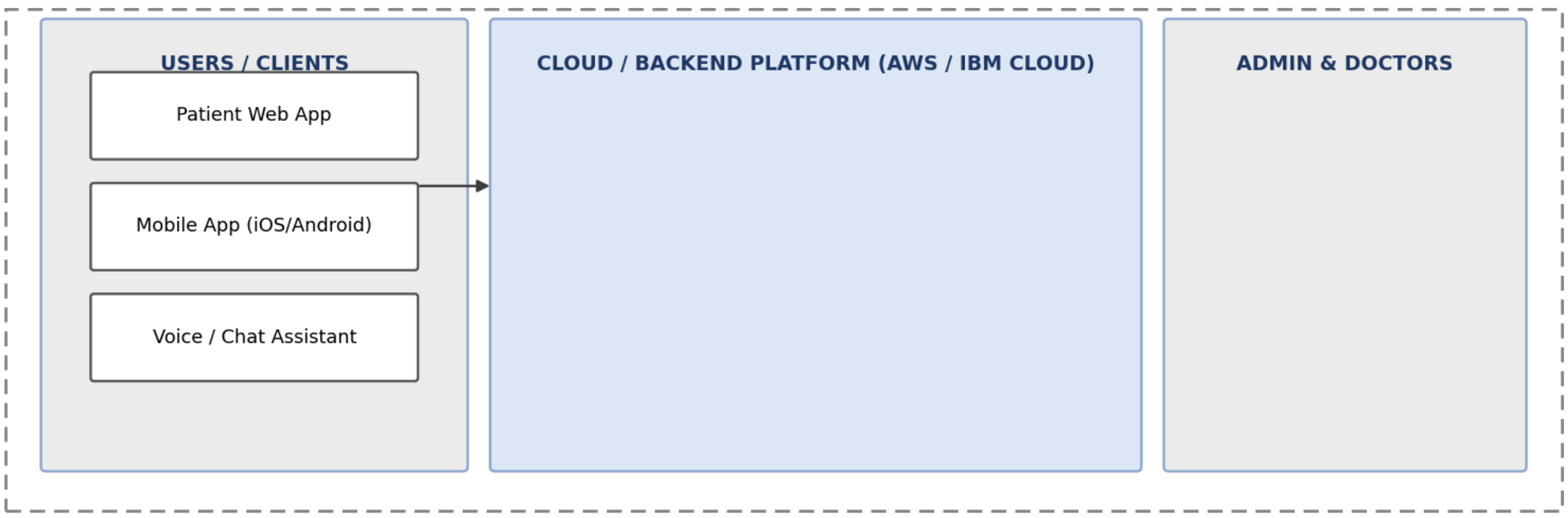
Student Name: Navendhiran C

Project Name: Book A Doctor

Maximum Marks: 4 Marks

Reference Architecture: Cloud-based Healthcare Scheduling & Telehealth Reference (AWS / IBM)

TECHNICAL ARCHITECTURE DIAGRAM



Guidelines Followed:

- Covers the complete application flow (Booking, Video Consultation, ML-based Triage, Notifications).
- Layers are clearly split into Client Tier, Cloud Backend, and Admin Portal.
- External integrations shown: Payment Gateways (Stripe), SMS/Video (Twilio/Agora), Government ID verification (Aadhaar API).
- Data storage and ML touchpoints indicated: managed Relational/NoSQL databases, Object Storage, and ML inference models.

TABLE-1: COMPONENTS & TECHNOLOGIES

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	**User Interface**	Patient web portal and mobile app for finding doctors, booking slots, and joining tele-consultations.	React.js, React Native, Tailwind CSS
2	**Application Logic-1**	Core service handling doctor schedule management, live slot booking logic, and the booking state machine.	Node.js (Express) Python (FastAPI)
3	**Application Logic-2**	AI chatbot assistant guiding voice&text appointment booking and initial patient screening.	IBM Watson Assistant Dialogflow
4	**Application Logic-3**	Tele-consultation engine running real-time WebRTC audio/video sessions between doctors and patients.	Twilio Programmable Video Agora SDK
5	**Database**	Primary store for structured data such as user profiles, doctor schedules, and transaction records.	PostgreSQL MySQL
6	**Cloud Database**	Managed NoSQL store for unstructured consultation notes, prescriptions, and chat logs.	MongoDB Atlas IBM Cloudant
7	**File Storage**	Encrypted object storage for diagnostic reports, medical records, and digital prescriptions.	AWS S3 IBM Cloud Object Storage
8	**External API-1**	Payment gateway API handling consultation fees via cards, UPI, and digital wallets.	Stripe API Razorpay API
9	**External API-2**	Identity and SMS verification API for patient/doctor checks and appointment reminders.	Aadhaar API Twilio SMS API
10	**Machine Learning Model**	Symptom checker and triage model suggesting the right medical specialty from reported symptoms.	Python, TensorFlow Scikit-Learn

11	**Infrastructure**	Containerized, auto-scaling deployment for handling peak booking load and secure API hosting.	Docker, Kubernetes (EKS IKS), NGINX

PROJECT DESIGN PHASE-II

Book A Doctor - Architectural Traits & References

Student Name: Deepan S R

TABLE-2: APPLICATION CHARACTERISTICS

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	**Open-Source Frameworks**	Relies on modular open-source libraries for backend services, frontend rendering, and background jobs like automated email/SMS reminders.	React.js, Node.js Express, FastAPI, Celery, Redis
2	**Security Implementations**	Follows HIPAA/GDPR health-data privacy rules closely. Uses end-to-end AES-256 encryption at rest, TLS 1.3 in transit, Role-Based Access Control (RBAC), and JWT-based authentication.	AES-256, TLS 1.3, Auth0 / JWT, OWASP Top 10 Protections, AWS WAF
3	**Scalable Architecture**	Built as stateless microservices so the booking engine, video streaming, and profile management can each scale out independently during demand spikes.	Kubernetes HPA (Horizontal Pod Autoscaler), Docker, NGINX Ingress
4	**Availability**	Designed for high availability with multi-region deployment, active-passive failover database clusters, and cloud load balancing targeting 99.9% uptime for urgent bookings.	AWS Elastic Load Balancing (ELB) / Cloudflare Traffic Manager, PostgreSQL Multi-AZ
5	**Performance Design**	In-memory caching of doctor availability calendars cuts DB queries by roughly 80%. A CDN speeds up static asset delivery and keeps API responses under 100ms.	Redis Cache, Cloudflare CDN, Database Indexing & Connection Pooling

PROJECT SUMMARY & WORKFLOW OVERVIEW

"Book A Doctor" is a full-featured digital health platform built to simplify how patients and physicians connect. It offers online appointment scheduling, ML-driven symptom-based specialty recommendations, secure video consultations, and digital prescriptions. Running on a scalable microservices architecture, the platform aims for near-zero downtime and fast response times even at peak booking hours, while enterprise-grade encryption and role-based access keep medical data protected in line with international healthcare regulations.

REFERENCES & DOCUMENTATION LINKS

- C4 Model for Software Architecture: https://c4model.com/
- IBM Developer Architecture Pattern - Online Order & Healthcare Processing: https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
- IBM Cloud Architecture Center: https://www.ibm.com/cloud/architecture
- AWS Reference Architecture Center: https://aws.amazon.com/architecture
- Guide to Technical Architecture Diagrams: https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d